

Singular Integral Operators

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1 Harmonic Analysis Facts

Theorem 1.1. *If T is a bounded linear transformation from $L^1(\mathbb{R}^n)$ to $L^1(\mathbb{R}^n)$, then T commutes with translations iff there exists a measure in $\mathcal{B}(\mathbb{R}^n)$ such that $T(f) = f * \mu$.*

Theorem 1.2. *If T is a bounded linear transformation from $L^2(\mathbb{R}^n)$ to $L^2(\mathbb{R}^n)$, then T commutes with translations if T is a multiplier operator, i.e. there exists bounded measurable $m(y)$ such that $(\hat{T}f)(y) = \hat{f}(y)m(y)$.*

2 Calderon-Zygmund Theory

Theorem 2.1. *Suppose a kernel $K(x)$ satisfies*

$$|K(x)| \leq \frac{B}{|x|^n} \quad |x| > 0, \quad (2.1)$$

$$\int_{R_1 < |x| < R_2} K(x) dx = 0 \quad 0 < R_1 < R_2 < \infty \quad (2.2)$$

and

$$\int_{|x| \geq 2|y|} |K(x-y) - K(x)| dx \leq B. \quad (2.3)$$

Then for $f \in L^1 \cap L^p(\mathbb{R}^n)$, $\epsilon > 0$ we define

$$T_\epsilon(f)(x) = \int_{|y| \geq \epsilon} f(x-y)K(y)dy.$$

Then have the following bound

$$\|T_\epsilon f\|_p \leq A_p \|f\|_p.$$

Moreover, $\lim_{\epsilon \rightarrow 0} T_\epsilon(f) = T(f)$ exists in L^p norm.

The strategy is to prove that T_ϵ is a weak (1,1) and strong (2,2) operator, with bounds independent of epsilon. Then apply Marcinkiewicz interpolation, and then check that bounds are preserved as $\epsilon \rightarrow 0$.

Remark 2.2. The condition that $K \in C^1(\mathbb{R}^n \setminus \{0\})$ and $|\nabla K(x)| \leq B/|x|^{n+1}$ implies (2.3).

2.1 L^2 to L^2 boundedness

Lemma 2.3. *Take K as above and define $K_\epsilon(x) = K(x)\mathbf{1}_{|x|\geq\epsilon}(x)$, which lies in L^2 . We have the estimates*

$$\sup_y |\hat{K}_\epsilon(y)| \leq CB$$

independently of ϵ .

Proof. It suffices to consider $\epsilon = 1$. $K_1(x)$ satisfies the same conditions as $K(x)$? The Fourier transform is given by

$$\begin{aligned} \hat{K}_1(y) &= \lim_{R \rightarrow \infty} \int_{|x| \leq R} e^{2\pi i x \cdot y} K_1(x) dx = I_1 + I_2 \\ &= \int_{|x| \leq \frac{1}{|y|}} e^{2\pi i x \cdot y} K_1(x) dx + \lim_{R \rightarrow \infty} \int_{\frac{1}{|y|} \leq |x| \leq R} e^{2\pi i x \cdot y} K_1(x) dx. \end{aligned}$$

Observe via the cancellation condition that

$$\begin{aligned} I_1 &= \int_{|x| \leq \frac{1}{|y|}} (e^{2\pi i x \cdot y} - 1) K_1(x) dx \\ |I_1| &\leq C|y| \int_{|x| \leq \frac{1}{|y|}} |x| |K_1(x)| dx \leq C|y| \int_{|x| \leq \frac{1}{|y|}} \frac{B|x|}{|x|^n} dx, \end{aligned}$$

so we switch to polar coordinates to see that the above is bounded by CB .

For I_2 , we define $z = \frac{1}{2} \frac{y}{|y|^2}$. Then $e^{2\pi i y \cdot z} = -1$, with $|z| = \frac{1}{2|y|}$. By a change of variables, we have

$$\int_{\mathbb{R}^n} K_1(x) e^{2\pi i x \cdot \xi} dx = \frac{1}{2} \int_{\mathbb{R}^n} [K_1(x) - K_1(x - z)] e^{2\pi i x \cdot \xi} dx$$

Some rearrangement tells us that

$$\begin{aligned} I_2 &= I_3 + I_4 \\ &= \frac{1}{2} \lim_{R \rightarrow \infty} \int_{\frac{1}{|y|} \leq |x| \leq R} [K_1(x) - K_1(x - z)] e^{2\pi i x \cdot \xi} dx - \frac{1}{2} \int_{|x| \leq \frac{1}{|y|}} [K_1(x) + K_1(x - z)] e^{2\pi i x \cdot \xi} dx. \end{aligned}$$

Then for I_3 observe that $|z| \leq \frac{1}{2}|x|$, so apply (2.3). By a change of variables, we rewrite I_4 as

$$\int_{|x| \leq \frac{1}{|y|}} K_1(x) e^{2\pi i x \cdot \xi} dx - \int_{|x+z| \leq \frac{1}{|y|}} K_1(x) e^{2\pi i x \cdot \xi} dx.$$

The integral is bounded in some spherical shell $\frac{1}{2|y|} \leq |x| \leq \frac{2}{|y|}$, so we can apply (2.1) with changes to spherical coordinates.

If T corresponds to kernel K then $\tau_{\epsilon^{-1}} T \tau_\epsilon$ corresponds to kernel $\epsilon^{-n} K(\epsilon^{-1}x)$. So let $K' = \epsilon^n K(\epsilon x)$. We can check that K' satisfies the conditions of the lemma with K'_1 satisfies $|\hat{K}'_1(y)| \leq CB$. But the Fourier transform of $K_\epsilon(x) = \epsilon^{-n} K'_1(\epsilon^{-1}x)$ is $\hat{K}_1(\epsilon y)$. $\square$

If the kernel has bounded Fourier transform, Plancherel gives

$$\|Tf\|_2 \leq B \|f\|_2$$

2.2 Weak L^1 to L^1 boundedness

Let $\alpha > 0$ be fixed. Our goal is to find a constant C such that

$$m\{Tf > \alpha\} \leq \frac{C}{\alpha} \int_{\mathbb{R}^n} |f(x)| dx.$$

First apply the Calderon-Zygmund decomposition, we can write $\mathbb{R}^n = F \cup \Omega$. Then define

$$g(x) := \begin{cases} f(x) & x \in F, \\ \int_{Q_j} f(x) dx & x \in Q_j^0. \end{cases}$$

Then take $b(x) = f(x) - g(x)$. Observe that $b(x)$ satisfies $b(x) = 0$ for $x \in F$ as well as

$$\int_{Q_j} b(x) dx = 0.$$

The triangle inequality tells us we can bound Tg and Tb separately since

$$m\{|Tf| > \alpha\} \leq m\{|Tg| > \frac{\alpha}{2}\} + m\{|Tb| > \frac{\alpha}{2}\}.$$

2.2.1 Estimate for Tg

We have

$$\begin{aligned} \|g\|_2^2 &= \int_F |g|^2 + \int_{\Omega} |g|^2 \\ &\leq \int_F \alpha |f| + m(\Omega)(2^n \alpha)^2 \\ &\leq \alpha \|f\|_1 + \frac{A}{\alpha} \|f\|_1 \cdot C^2 \alpha^2 \leq \frac{C}{\alpha} \|f\|_1. \end{aligned}$$

But from the $L^2 \rightarrow L^2$ bound, we can apply Tchebychev for

$$m\{|Tg| > \frac{\alpha}{2}\} \leq \frac{C^2}{\alpha^2} \|g\|_2^2.$$

2.2.2 Estimate for Tb

Here, we want the estimate for the integral of Tb over the good set F .

We can write $b_j(x) = b(x)\mathbf{1}_{Q_j}(x)$. We now expand the cubes each by $2\sqrt{n}$ times, denoted Q_j^* . Take F^* as the complement of the union of the expanded cubes. For $x \in F^*$, we can check that $|x - y^j| \geq 2|y - y^j|$ for all $y \in Q_j$, where y^j is the center of the cube. Then

$$\begin{aligned}
Tb_j(x) &= \int_{Q_j} K(x - y)b_j(y)dy \\
&= \int_{Q_j} [K(x - y) - K(x - y^j)]b_j(y)dy, \\
|Tb(x)| &\leq \sum_j \int_{Q_j} |K(x - y) - K(x - y^j)||b(y)|dy, \\
\int_{F^*} |Tb(x)|dx &\leq \int_{F^*} \sum_j \int_{Q_j} |K(x - y) - K(x - y^j)||b(y)|dydx \\
&\leq \sum_j \int_{Q_j} |b(y)| \int_{F^*} |K(x - y) - K(x - y^j)|dx dy \quad (\text{Fubini's}), \\
&\leq \sum_j \int_{Q_j} |b(y)| \int_{|x'| \geq 2|y'|} |K(x' - y') - K(x')|dx' dy \\
&\leq B \sum_j \int_{Q_j} |b(y)|dy \leq C \|f\|_1.
\end{aligned}$$

We also recall that $m(\Omega) \leq \frac{C}{\alpha} \|f\|_1$.

2.3 Duality

How do we do this approximation if K does not lie in L^2 ? Answer: We can sidestep this because K_ϵ lies in L^2 .

Recall that if $\psi \in L^1_{\text{loc}}$ and

$$\sup \left\{ \left| \int \psi \varphi dx \right| : \varphi \in C_c, \|\varphi\|_q \leq 1 \right\} = A < \infty,$$

then $\|\psi\|_{L^p} = A$. We can write the double integral as

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} K(x - y)f(y)\varphi(x)dx dy = \int f(y) \left(\int K(x - y)\varphi(y)dx \right) dy.$$

Replacing with the kernel $K(-x)$, we know that $\int K(x - y)\varphi(y)dx$ belongs to L^q for $1 < q < 2$. Then apply Hölder's.

2.4 From T_ϵ to T

Consider $f \in C_c^1(\mathbb{R}^n)$. Then by the cancellation condition

$$T_\epsilon(f_1)(x) = \int_{|y| \geq 1} K(y) f_1(x-y) dy + \int_{1 \geq |y| \geq \epsilon} K(y) [f_1(x-y) - f_1(x)] dy.$$

The first integral lies in L^p , second integral converges uniformly to 0. We can write arbitrary $f \in L^p$ as $f = f_1 + f_2$, where $f_1 \in C_c^1$ and $\|f_2\|_p$ is small.

What does apply basic inequality for f_2 in place of f mean?

3 SIOs which commute with dilations

If $\tau_{\epsilon^{-1}} T \tau_\epsilon = T$, then we are back to the requirement $K(\epsilon x) = \epsilon^{-n} K(x)$, where K is homogeneous of degree $-n$. Then

$$K(x) = \frac{\Omega(x)}{|x|^n},$$

where Ω is homogeneous of degree 0.

Remark 3.1. It suffices to consider Ω which satisfies the following smoothness and cancellation conditions

$$\sup\{|\Omega(x) - \Omega(x')| : |x - x'| \leq \delta, |x| = |x'| = 1\} = \omega(\delta) \implies \int_0^1 \frac{\omega(\delta)}{\delta} d\delta < \infty, \quad (3.1)$$

$$\int_{S^{n-1}} \Omega(\sigma) d\sigma = 0. \quad (3.2)$$

Why does (3.3) suffice?

Theorem 3.2. *Let Ω homogeneous of degree 0 satisfying the above two properties. Let*

$$T_\epsilon(f)(x) = \int_{|y| > \epsilon} \frac{\Omega(y)}{|y|^n} f(x-y) dy.$$

Then T_ϵ is bounded from L^p to L^p . The limit in the L^p norm exists (call it T), and T satisfies the same bounds.

In addition to convergence in L^p norm, we can also get convergence almost everywhere, with the help of the maximal function.

Theorem 3.3. *Take Ω as above. For $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$, the limit $\lim_{\epsilon \rightarrow 0} T_\epsilon(f)(x)$ exists for almost everywhere. Define the associated maximal function*

$$T^*(f)(x) = \sup_{\epsilon > 0} |T_\epsilon(f)(x)|.$$

The result follows from the fact that T^ is of weak type $(1, 1)$ and strong type (p, p) .*

Proof. Proof of (p, p) is deferred to latter chapter.

Again given $\alpha > 0$, we will consider a splitting of a function $f = g + b$. Take $x \in (Q_j^*)^c$. Suppose that there exists some $y \in Q_j$ such that $|x - y| = \epsilon$. Then there exists γ_n and γ'_n such that for every $y' \in Q_j$,

$$\gamma'_n \epsilon \leq |x - y| \leq \gamma_n \epsilon.$$

We claim that if $x \in F^*$,

$$\sup_{\epsilon > 0} |T_\epsilon(b(x))| \leq \sum_j \int_{Q_j} |K(x - y) - K(x - y^j)| |b(y)| dy + C \sup_{r \rightarrow 0} \int_{B(x, r)} |b(y)| dy.$$

By definition,

$$T_\epsilon b(x) = \sum_j \int_{Q_j} K_\epsilon(x - y) b(y) dy.$$

There are three different kinds of Q_j :

1. For all $y \in Q_j$, we have $|x - y| < \epsilon$. This term vanishes.
2. For all $y \in Q_j$, we have $|x - y| > \epsilon$. This appears as the first term.
3. There exists $y \in Q_j$ such that $|x - y| = \epsilon$. Use the above bounds, and estimate over $B(x, \gamma_n \epsilon)$.

Now define

$$\Lambda(f)(x) = \left| \limsup_{\epsilon \rightarrow 0} T_\epsilon(f)(x) - \liminf_{\epsilon \rightarrow 0} T_\epsilon(f)(x) \right| \leq 2(T^*f)(x).$$

□

Hypothesis 3.4. *The Calderón-Zygmund theorem gives us bounds on certain singular integral operators. Can we in fact describe them as Fourier multipliers?*

Proposition 3.5. *Suppose Ω is homogeneous of degree 0, and suppose that Ω satisfies the following cancellation and smoothness conditions*

$$\sup\{|\Omega(x) - \Omega(x')| : |x - x'| \leq \delta, |x| = |x'| = 1\} = \omega(\delta) \implies \int_0^1 \frac{\omega(\delta)}{\delta} d\delta < \infty, \quad (3.3)$$

$$\int_{S^{n-1}} \Omega(\sigma) d\sigma = 0. \quad (3.4)$$

Let Tf be the convolution of $\Omega(x)/|x|^n$ with $f(x)$, defined in the principal value sense.

If $f \in L^2(\mathbb{R}^n)$, then the Fourier transforms of f and Tf are related by $\widehat{Tf} = m\hat{f}$, where m is a homogeneous function of degree 0. Explicitly, we have

$$m(x) = \int_{S^{n-1}} \left[\frac{\pi i}{2} \operatorname{sgn}(x \cdot y) + \log \left(\frac{1}{|x \cdot y|} \right) \right] \Omega(y) d\sigma(y). \quad (3.5)$$

Proof. Since T is bounded and commutes with translations, we can write T as a multiplier operator. Moreover, such an operator commutes with dilations, so the multiplier is homogeneous of degree 0. How do we express the multiplier in terms of the kernel? Take

$$K_{\epsilon, \eta}(x) = \begin{cases} \frac{\Omega(x)}{|x|^n} & \epsilon \leq |x| \leq \eta, \\ 0 & \text{otherwise.} \end{cases}$$

We can check immediately that $K_{\epsilon, \eta} \in L^1$ (though not necessarily uniformly). Moreover, if $f \in L^2(\mathbb{R}^n)$, then $\widehat{K_{\epsilon, \eta} * f} = \hat{K}_{\epsilon, \eta} \hat{f}$.

We claim the following

1. $\sup_{\epsilon, \eta} |\widehat{K_{\epsilon, \eta}}(y)| \leq A$.
2. If $x \neq 0$, then the limit as $\epsilon \rightarrow 0, \eta \rightarrow \infty$ of $\hat{K}_{\epsilon, \eta}(x) = m(x)$.

To this end, we write $x = Rx', y = ry'$ in polar coordinates. Consider the following integral

$$I_{\epsilon, \eta}(x', y') := \int_{\epsilon}^{\eta} \frac{1}{r} [\exp(2\pi i Rr x' \cdot y') - \cos(2\pi Rr)] dr.$$

With some calculus, we get

$$\begin{aligned} \operatorname{Im}(I_{\epsilon, \eta}(x', y')) &= \int_{\epsilon}^{\eta} \frac{\sin[2\pi Rr(x' \cdot y')]}{r} dr \rightarrow \frac{\pi}{2} \operatorname{sgn}(x' \cdot y'), \\ \operatorname{Re}(I_{\epsilon, \eta}(x', y')) &= \int_{\epsilon}^{\eta} \frac{\cos[2\pi Rr(x' \cdot y')] - \cos(2\pi Rr)}{r} dr \rightarrow \cos 0 \log \frac{1}{|x' \cdot y'|}, \end{aligned}$$

since

$$\int_0^\infty \frac{h(\lambda r) - h(\mu r)}{r} dr = h(0) \log \frac{\mu}{\lambda}.$$

Combining the real and imaginary parts, we have

$$I(x', y') \rightarrow \log \frac{1}{|x' \cdot y'|} + i \frac{\pi}{2} \operatorname{sgn}(x' \cdot y').$$

Rewriting in spherical coordinates, we can express the Fourier transform of the kernel as

$$\begin{aligned} \hat{K}_{\epsilon, \eta}(x) &= \int_{\mathbb{S}^{n-1}} \int_\epsilon^\eta e^{2\pi i R r x' \cdot y'} \Omega(y') \frac{dr}{r} d\sigma(y') \\ &= \int_{\mathbb{S}^{n-1}} I_{\epsilon, \eta}(x', y') \Omega(y') d\sigma(y') \end{aligned}$$

with the help of the cancellation property. But we can just take the norm of the real and imaginary parts of $I_{\epsilon, \eta}$ for the uniform bound

$$|\hat{K}_{\epsilon, \eta}(x)| \leq A \int_{\mathbb{S}^{n-1}} [1 + \log \frac{1}{|x' \cdot y'|}] |\Omega(y')| d\sigma(y').$$

Then apply DCT.

To show convergence to T , first take ϵ fixed and take $\eta \rightarrow \infty$. Then consider $\epsilon \rightarrow 0$. $\square$

Remark 3.6. No boundedness on L^1, L^∞ , take Hilbert transform of characteristic function of interval (a, b) .

4 Hilbert and Riesz transforms

Definition 4.1. The Hilbert transform is given by

$$Hf(x) := \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \int_{|y| \geq \epsilon} \frac{f(x-y)}{y} dy.$$

It turns out that the Hilbert transform satisfies the properties of the singular integral operators for which we proved the Calderón-Zygmund theorems. Here, we have

$$K(x) = \frac{1}{\pi x}, \Omega(x) = \frac{1}{\pi} \operatorname{sgn}(x) = \frac{1}{\pi} \frac{x}{|x|}.$$

Then $\widehat{Hf} = m\hat{f}$, with $m(x) = i \operatorname{sgn}(x)$. Then $H^2 = -I$.

Proposition 4.2. Suppose T is a bounded operator on $L^2(\mathbb{R}^1)$ which

1. commutes with translations,
2. commutes with positive dilations,
3. anticommutes with reflection.

Then T is a multiple of the Hilbert transform.

Proof. Since T commutes with translations, we can write $\widehat{Tf} = m\hat{f}$. To simplify notation, we write $\mathcal{F}f = \hat{f}$. Then $\mathcal{F}T = m\mathcal{F}$.

We recall the effect of a Fourier transform on dilation, given by

$$\begin{aligned} (\mathcal{F}\tau_\delta f)(y) &= \int e^{2\pi ixy} f(\delta x) dx \\ &= |\delta|^{-1} \int e^{2\pi ixy/\delta} f(x) dx = |\delta|^{-1} \tau_{\delta^{-1}} \mathcal{F}. \end{aligned}$$

Now our remaining assumptions imply that $T\tau_\delta = \text{sgn}(\delta)\tau_\delta T$. We have

$$\begin{aligned} \tau_\delta m &= \tau_\delta (\mathcal{F}T\mathcal{F}^{-1}) = |\delta|^{-1} \mathcal{F}\tau_{\delta^{-1}} T \mathcal{F}^{-1} \\ &= \delta^{-1} \mathcal{F}T\tau_{\delta^{-1}} \mathcal{F}^{-1} \\ &= \text{sgn}(\delta) \mathcal{F}T\mathcal{F}^{-1} \tau_\delta = \text{sgn}(\delta) m \tau_\delta. \end{aligned}$$

Specifically, we have

$$m(\delta y) \hat{f}(\delta y) = \text{sgn}(\delta) m(y) \hat{f}(\delta y).$$

Then $m(y)$ is a constant multiple of $\text{sgn}(y)$. □

Remark 4.3. If T is a bounded linear operator on $L^2(\mathbb{R}^1)$ that commutes with translations and all dilations, then its Fourier multiplier is a constant. Then T is a constant multiple of the identity.

Let ρ denote a rotation. We define $\rho(f)(x) = f(\rho^{-1}x)$. We can verify $\mathcal{F}\rho = \rho\mathcal{F}$.

Lemma 4.4. *Let $m(x) = (m_1(x), m_2(x), \dots, m_n(x))$ be an n -tuple of functions on $\mathbb{R}^n$. Suppose that*

1. m is homogeneous of degree 0,
2. m transforms like a vector, i.e.

$$\rho(m)(x) = m(\rho^{-1}x) = \rho(m(x)), \quad m_j(\rho^{-1}x) = \sum_k \rho_{jk} m_k(x). \quad (4.1)$$

Here we take the induced action of ρ on a function.

Then the function m takes the form

$$m(x) = c \frac{x}{|x|}, \quad m_j(x) = c \frac{x_j}{|x|}.$$

Proof. It suffices to consider x on the unit sphere. Let $\{e_i : 1 \leq i \leq n\}$ denote the standard basis. Set $c = m_1(e_1)$.

Let ρ be a rotation that fixes e_1 . For $2 \leq j \leq n$, we have $\rho_{j1} = 0$ which means

$$m_j(e_1) = \sum_{k=2}^n \rho_{jk} m_k(e_1).$$

So the $n-1$ dimensional vector $(m_2(e_1), m_3(e_1), \dots, m_n(e_1))$ is left fixed by all the rotations on this $n-1$ dimensional space orthogonal to e_1 . So $m_2(e_1) = m_3(e_1) = \dots = m_n(e_1) = 0$. We obtain

$$m_j(\rho^{-1}e_1) = \rho_{j1}m_1(e_1) = c\rho_{j1}.$$

If $\rho^{-1}e_1 = x$, then $\rho_{j1} = x_j$. Then $m_j(x) = cx_j$. □

Definition 4.5. Let $f \in L^p(\mathbb{R}^n)$, $1 \leq p < \infty$. We define

$$R_j(f)(x) = \lim_{\epsilon \rightarrow 0} c_n \int_{|y| \geq \epsilon} \frac{y_j}{|y|^{n+1}} f(x-y) dy, \quad c_n = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\pi^{\frac{n+1}{2}}}$$

The associated kernel is

$$K_j(x) = c_n \frac{x_j}{|x|^{n+1}}, \quad \Omega_j(x) = c_n \frac{x_j}{|x|}.$$

Observe that the mapping from kernel Ω to multipliers m commutes with rotations. Moreover, the kernels satisfy the transformation law (4.1), which means that the multipliers also satisfy the transformation law.

But the m_j 's are homogeneous of degree 0, so the lemma shows that $m_j(x) = cx_j/|x|$. Notice that

$$c = \int_{\mathbb{S}^{n-1}} \left(\frac{\pi i}{2} \operatorname{sgn}(y_1) + \log \left| \frac{1}{y_1} \right| \right) \cdot c_n \frac{y_j}{|y|} d\sigma(y) = i$$

In this case we just need to check that

$$\frac{2\Gamma\left(\frac{n+1}{2}\right)}{\pi^{(n+1)/2}} = \int_{\mathbb{S}^{n-1}} |\cos \theta| d\sigma(y). \quad (4.2)$$

In other words, we have

$$\widehat{R_j f}(y) = i \frac{y_j}{|y|} \hat{f}(y).$$

The Riesz operators obey

$$\rho^{-1} R_j \rho f = \sum_k \rho_{jk} R_k f.$$

If we denote $\hat{R}_j = m_j$, then

$$\rho(m_j \rho^{-1}(f)) = \sum_k \rho_{jk} m_k f \iff m_j(\rho^{-1}x) = \sum_k \rho_{jk} m_k(x).$$

Proposition 4.6. *Let $T = (T_1, T_2, \dots, T_n)$ be an n -tuple of bounded transformations on $L^2(\mathbb{R}^n)$. Suppose*

1. *Each T_j commutes with translation*
2. *Each T_j commutes with dilations*
3. *For every rotation ρ , $\rho^{-1} T_j \rho f = \sum_k \rho_{jk} T_k f$.*

Then the T_j 's are a constant multiple of the Riesz transforms.

4.1 Applications of Riesz transforms

Proposition 4.7. *Suppose $f \in C_C^2$. Then for $1 < p < \infty$.*

$$\left\| \frac{\partial^2 f}{\partial x_j \partial x_k} \right\|_p \leq A_p \|\Delta f\|_p. \quad (4.3)$$

Proof. We want to apply the identity

$$\frac{\partial^2 f}{\partial x_j \partial x_k} = -R_j R_k \Delta f.$$

Recall the action of the Fourier transform of derivatives

$$\hat{f}(y) = \int e^{2\pi i x \cdot y} f(x) dx \implies \frac{\partial \hat{f}}{\partial x_j}(y) = -2\pi i y_j \hat{f}(y).$$

Then

$$\begin{aligned} \widehat{\frac{\partial^2 f}{\partial x_j \partial x_k}}(y) &= -4\pi^2 y_j y_k \hat{f}(y) \\ &= -\frac{i y_j}{|y|} \frac{i y_k}{|y|} (-4\pi |y|^2) \hat{f}(y) = -\widehat{R_j R_k \Delta f}(y). \end{aligned}$$

□

Proposition 4.8. *Suppose $f \in C^1(\mathbb{R}^2)$ with compact support. Then for $1 < p < \infty$,*

$$\left\| \frac{\partial f}{\partial x_1} \right\|_p + \left\| \frac{\partial f}{\partial x_2} \right\|_p \leq A_p \left\| \frac{\partial f}{\partial x_1} + i \frac{\partial f}{\partial x_2} \right\|_p.$$

The identity used is

$$\frac{\partial f}{\partial x_j} = -R_j(R_1 - iR_2) \left(\frac{\partial f}{\partial x_1} + i \frac{\partial f}{\partial x_2} \right).$$

5 Poisson integrals

Recall the Dirichlet problem for the Laplace equation: We restrict ourself to $\mathbb{R}_+^{n+1}$. Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we want to find a harmonic function $u(x, y)$ on $\mathbb{R}_+^{n+1}$ whose boundary values on $\mathbb{R}^n$ are $f(x)$.

Here is a solution with L^2 theory. Let $f \in L^2$. Consider

$$u(x, y) = \int \hat{f}(t) e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt.$$

This integral converges absolutely, and can be differentiated. Then check

$$\Delta u = \frac{\partial^2 u}{\partial y^2} + \sum_{k=1}^n \frac{\partial^2 u}{\partial x_k^2} = 0,$$

because $e^{2\pi i t \cdot x} e^{-2\pi |t| y}$ satisfies this property. We have L^2 convergence of $u(x, y)$ to $f(x)$ as $y \rightarrow 0$.

Definition 5.1. Define the Poisson kernel $P_y(x)$ by

$$P_y(x) = \int e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt.$$

Then we can write $u(x, y)$ as a convolution in the x variable called the Poisson integral

$$u(x, y) = (P_y * f)(x) = \int P_y(t) f(x - t) dt$$

Proposition 5.2. *The explicit expression of the Poisson kernel is*

$$P_y(x) = \frac{c_n y}{(|x|^2 + y^2)^{\frac{n+1}{2}}},$$

with c_n as in the Riesz transform.

We can describe the boundary behavior of Poisson integrals as follows.

Theorem 5.3. Let $f \in L^p(\mathbb{R}^n)$, $1 \leq p \leq \infty$. Let $u(x, y)$ be its associated Poisson integral. Then

1. $\sup_{y>0} |u(x, y)| \leq Mf(x)$, where Mf is the maximal function.
2. $\lim_{y \rightarrow 0} u(x, y) = f(x)$ for almost every x .
3. If $p < \infty$, then $u(x, y)$ converges to $f(x)$ in the $L^p(\mathbb{R}^n)$ norm as $y \rightarrow 0$.

Theorem 5.4. [Approximations to the identity] Let $\varphi \in L^1(\mathbb{R}^n)$, and set $\varphi_\epsilon(x) = \epsilon^{-n} \varphi(\epsilon^{-1}x)$. Suppose that the least decreasing radial majorant of φ is integral, i.e.

$$\psi(x) = \sup_{|y| \geq |x|} |\varphi(y)| \text{ satisfies } \int \psi(x) dx = A < \infty.$$

With the same A ,

1. $\sup_{\epsilon>0} |(f * \varphi_\epsilon)(x)| \leq AMf(x)$, for $f \in L^p(\mathbb{R}^n)$, $1 \leq p \leq \infty$.
2. If φ has integral one, then $\lim_{\epsilon \rightarrow 0} (f * \varphi_\epsilon)(x) = f(x)$ almost everywhere.
3. If $p < \infty$, then $\|f * \varphi_\epsilon - f\|_p \rightarrow 0$ as $\epsilon \rightarrow 0$.

Proof. To prove (3), it suffices to take φ integrable. Recall the L^p norm is continuous with respect to translation. If f_1 is continuous with compact support, we in fact have uniform convergence of $f_1(x - y)$ to $f_1(x)$. Otherwise write $f = f_1 + f_2$, where $\|f_2\|_p \leq \delta$. Now

$$\begin{aligned} \|f(x - y) - f(x)\|_p &\leq \|f_1(x - y) - f_1(x)\|_p + \|f_2(x - y) - f_2(x)\|_p, \\ \|f_2(x - y) - f_2(x)\|_p &\leq 2\delta, \end{aligned}$$

so $\|f(x - y) - f(x)\|_p \rightarrow 0$. We can write with Fubini's

$$\begin{aligned} f * \varphi_\epsilon - f &= \int [f(x - y) - f(x)] \varphi_\epsilon(y) dy \\ \|f * \varphi_\epsilon - f\|_p &\leq \int \|f(x - y) - f(x)\|_{L_x^p} |\varphi_\epsilon(y)| dy \\ &= \int \|f(x - \epsilon y) - f(x)\|_{L_x^p} |\varphi(y)| dy \end{aligned}$$

which converges to zero by DCT.

To prove (1), write $\psi(r) = \psi(x)$, since ψ is radial. We claim that $r^n \psi(r) \rightarrow 0$ as $r \rightarrow 0, \infty$. Indeed, we can write

$$\int_{\frac{r}{2} \leq |x| \leq r} \psi(x) dx \geq \psi(r) \int_{\frac{r}{2} \leq |x| \leq r} dx = \psi(r) c r^n,$$

then we apply the fact that $\psi \in L^1$, $\psi(r)$ is decreasing (and absolute continuity of the integral?).

We now want to show that

$$(f * \psi_\epsilon)(x) \leq A(Mf)(x).$$

By translation and dilation invariance, it suffices to show that

$$(f * \psi)(0) \leq A(Mf)(0).$$

Write

$$\begin{aligned}\lambda(r) &= \int_{\mathbb{S}^{n-1}} f(rx) d\sigma(x), \\ \Lambda(r) &= \int_{|x| \leq r} f(x) dx = \int_0^r \lambda(t)^{n-1} dt,\end{aligned}$$

by polar coordinates. Then

$$\begin{aligned}(f * \psi)(0) &= \int f(x) \psi(x) dx = \int_0^\infty \lambda(r) \psi(r) r^{n-1} dr \\ &= \lim_{\epsilon \rightarrow 0, N \rightarrow \infty} \int_\epsilon^N \lambda(r) \psi(r) r^{n-1} dr = - \lim_{\epsilon \rightarrow 0, N \rightarrow \infty} \int_\epsilon^N \Lambda(r) d\psi(r).\end{aligned}$$

We have an error of the form $\Lambda(N)\psi(N) - \Lambda(\epsilon)\psi(\epsilon)$, but we can check that

$$\Lambda(r) \leq |B(1)| r^n Mf(0).$$

Therefore

$$f * \psi(0) \leq VMf(0) \int_0^\infty r^n d(-\psi(r)).$$

To prove (2), if $f \in L^p$, $1 \leq p < \infty$, proof is analogous to Lebesgue Differentiation Thm. Now take $p = \infty$. Given any ball B , we want to show that

$$\lim_{\epsilon \rightarrow 0} (f * \varphi_\epsilon)(x) = f(x)$$

for almost every $x \in B$. Let B_1 be a different ball that strictly contains B and δ be the distance from B to the complement of B_1 .

Take $f_1(x) = f(x)\mathbf{1}_B(x)$, $f(x) = f_1(x) + f_2(x)$. We have $f_1 \in L^1$. For $x \in B$,

$$\begin{aligned}|(f_2 * \varphi_\epsilon)(x)| &= \left| \int f_2(x-y) \varphi_\epsilon(y) dy \right| \leq \int_{|y| \geq \delta} |f_2(x-y)| \varphi_\epsilon(y) dy \\ &\leq \|f\|_\infty \int_{|y| \geq \frac{\delta}{\epsilon}} |\varphi(y)| dy \rightarrow 0\end{aligned}$$

as $\epsilon \rightarrow 0$. □

5.1 Conjugate harmonic functions

There is an interesting relation between Riesz transform and the theory of harmonic functions.

Theorem 5.5. *Let $f, f_i \in L^2(\mathbb{R}^n)$, with their respective Poisson integrals*

$$u_0(x, y) = P_y * f, u_i(x, y) = P_y * f_i.$$

Then $f_j = R_j(f)$ iff the following generalized Cauchy-Riemann equations hold

$$\sum_{j=0}^n \frac{\partial u_j}{\partial x_j} = 0, \quad \frac{\partial u_j}{\partial x_k} = \frac{\partial u_k}{\partial x_j}.$$

Proof. ($\implies$) Since $f_j = R_j(f)$, we know that $\hat{f}_j(t) = \frac{it_j}{|t|} \hat{f}(t)$. Then the formula for the Poisson integral is

$$u_j(x, y) = \int \hat{f}(t) \frac{it_j}{|t|} e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt.$$

Then just differentiate under the integral sign.

($\impliedby$) Consider the formula for the Poisson integral. We have

$$-2\pi i t_j \hat{f}_0(t) e^{-2\pi |t| y} = -2\pi |t| \hat{f}_j(t) e^{-2\pi |t| y},$$

so $\hat{f}_j(t) = \frac{it_j}{|t|} \hat{f}_0(t)$ and $f - J = R_j(f)$. □

5.2 L^p bounds on maximal singular operator

Lemma 5.6. *If $T^*f(x) = \sup_{\epsilon > 0} |T_\epsilon(f)|(x)$, then*

$$\|T^*f\|_p \leq A_p \|f\|_p, \quad 1 < p < \infty.$$

Proof. We already proved existence of $T(f)$ as limit as $\epsilon \rightarrow 0$ in the L^p norm. We want to show that

$$T^*(f)(x) \leq M(Tf)(x) + CM(f)(x).$$

Let φ be a smooth non-negative function on $\mathbb{R}^n$, supported in unit ball, with integral equal to one, radial and decreasing in $|x|$. Consider

$$K_\epsilon(x) = \begin{cases} \frac{\Omega(x)}{|x|^n} & |x| \geq \epsilon, \\ 0 & |x| < \epsilon. \end{cases}$$

Then we define $\Phi = \varphi * K - K_1$.

We claim that the smallest decreasing radial majorant of Φ is integrable.

- If $|x| < 1$, then $\Phi = \varphi * K$ and we can write

$$\Phi = \int K(y)[\varphi(x-y) - \varphi(x)]dy$$

which is bounded due to smoothness of φ .

- If $1 \leq |x| \leq 2$, then $\Phi(x) = K * \varphi - K(x)$ is also bounded.
- When $|x| \geq 2$,

$$\Phi(x) = \int_{\mathbb{R}^n} K(x-y)\varphi(y)dy - K(x) = \int_{|y| \leq 1} [K(x-y) - K(x)]\varphi(y)dy$$

so

$$|\Phi(x)| \leq C' \frac{\omega(c/|x|)}{|x|^n}.$$

Observe too that

$$\varphi_\epsilon * K - K_\epsilon = \Phi_\epsilon.$$

We claim now that for any $f \in L^p(\mathbb{R}^n)$, we have

$$(\varphi_\epsilon * K) * f(x) = T(f) * \varphi_\epsilon(x)$$

We conclude $T_\epsilon(f) = (Tf) * \varphi_\epsilon - f * \Phi_\epsilon$, so we can apply [Theorem 5.4](#). □

6 Higher Riesz transforms and spherical harmonics

Definition 6.1. $\mathcal{H}_k$ is the linear space of homogeneous polynomials of degree k , also known as the solid spherical harmonic of degree k . This space has inner product

$$(P, Q) = \int_{\mathbb{S}^{n-1}} P(x) \overline{Q(x)} d\sigma(x).$$

Proposition 6.2. *The space $\{\mathcal{H}_k\}_{k=0}^\infty$ are orthogonal.*

Proof. If $P \in \mathcal{H}_k, Q \in \mathcal{H}_j$ then

$$\begin{aligned} (k-j) \int_{\mathbb{S}^{n-1}} P \overline{Q} d\sigma(x) &= \int_{\mathbb{S}^{n-1}} \left(\overline{Q} \frac{\partial P}{\partial \nu} - P \frac{\partial \overline{Q}}{\partial \nu} \right) d\sigma(x) \\ &= \int_{B_1} (\overline{Q} \Delta P - P \Delta \overline{Q}) dx = 0, \end{aligned}$$

because P and Q are both harmonic. □

Proposition 6.3. *Suppose P is homogeneous of degree k . Then*

$$P = P_1 + |x|^2 P_2,$$

where P_1 is homogeneous of degree k , harmonic and P_2 is homogeneous of degree $k-2$.

Proof. Iterate the previous proposition for

$$P(x) = P_1(x) + |x|^2 P_2(x) + |x|^4 P_3(x) + \cdots =$$

□

Proposition 6.4. *Let H_k denote the linear space of restrictions of $\mathcal{H}_k$ to the unit sphere, also known as the surface spherical harmonics of degree k . Then in the sense of Hilbert spaces,*

$$L^2(\mathbb{S}^{n-1}) = \sum_{k=0}^{\infty} H_k.$$

Proposition 6.5. *Let f be written as*

$$f(x) = \sum_{k=0}^{\infty} Y_k(x).$$

Then f is smooth on $\mathbb{S}^{n-1}$ if and only if

$$\int_{\mathbb{S}^{n-1}} |Y_k(x)|^2 d\sigma(x) = O(k^{-N}). \tag{6.1}$$

Theorem 6.6. Suppose $P_k(x)$ is a homogeneous harmonic polynomial of degree k . Then

$$\mathcal{F}(P_k(x)e^{-\pi|x|^2}) = i^k P_k(y)e^{-\pi|y|^2}.$$

Proof. (What's wrong with the following approach? We don't use harmonicity) We want to show that

$$\int P_k(x) \exp(-\pi|x|^2 + 2\pi i x \cdot y) dx = i^k P_k(y) e^{-\pi|y|^2}. \quad (6.2)$$

The Fourier transform of a Gaussian is a Gaussian, so

$$\int \exp[-\pi|x|^2 + 2\pi i x \cdot y] dx = \exp(-\pi|y|^2).$$

Now apply the operator $P_k(\partial_y)$ to both sides for

$$\int P_k(2\pi i x) \exp[-\pi|x|^2 + 2\pi i x \cdot y] dx = P_k(-2\pi y) \exp(-\pi|y|^2).$$

□

Theorem 6.7. Let $P_k(x)$ be a homogeneous harmonic polynomial of degree k . Then the multiplier corresponding to the convolution operator with the kernel $P_k(x)/|x|^{k+n}$ is

$$\gamma_k \frac{P_k(x)}{|x|^k}, \text{ where } \gamma_k = i^k \pi^{n/2} \frac{\Gamma(k/2)}{\Gamma((k+n)/2)}.$$

Lemma 6.8. For all $k \in \mathbb{N}$, $0 < \alpha < n$, and $\varphi \in \mathcal{S}(\mathbb{R}^n)$ we have

$$\int \frac{P_k(x)}{|x|^{k+n-\alpha}} \hat{\varphi}(x) dx = \gamma_{k,\alpha} \int \frac{P_k(x)}{|x|^{k+\alpha}} \varphi(x) dx.$$

Proof of lemma.

□

Proof. We claim that

$$\lim_{\alpha \rightarrow 0^+} \int \frac{P_k(x)}{|x|^{k+n-\alpha}} \hat{\varphi}(x) dx = \lim_{\epsilon \rightarrow 0} \int_{|x| \geq \epsilon} \frac{P_k(x)}{|x|^{k+n}} \hat{\varphi}(x) dx. \quad (6.3)$$

The left hand side is

$$\int \frac{P_k(x)}{|x|^{k+n-\alpha}} \hat{\varphi}(x) dx = \int_{|x| \leq 1} \frac{P_k(x)}{|x|^{k+n-\alpha}} [\hat{\varphi}(x) - \hat{\varphi}(0)] dx + \int_{|x| \geq 1} \frac{P_k(x)}{|x|^{k+n-\alpha}} \hat{\varphi}(x) dx.$$

How do we pass to the limit as $\alpha \rightarrow 0$? Somehow

$$\lim_{\alpha \rightarrow 0} \int_{|x| \leq 1} \frac{P_k(x)}{|x|^{k+n-\alpha}} [\hat{\varphi}(x) - \hat{\varphi}(0)] dx = \lim_{\epsilon \leq |x| \leq 1} \frac{P_k(x)}{|x|^{k+n}} \hat{\varphi}(x) dx$$

In any case let f be sufficiently smooth with compact support. Set $f(x - y) = \hat{\varphi}(y)$, so $\varphi(y) = \hat{f}(y)e^{-2\pi i x \cdot y}$, so we get

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{|y| \geq \epsilon} \frac{P_k(y)}{|y|^{k+n}} f(x - y) dy &= \gamma_k \int \frac{P_k(y)}{|y|^k} \hat{f}(y) e^{-2\pi i x \cdot y} dy \\ &= \int m(y) \hat{f}(y) e^{-2\pi i x \cdot y} dy \end{aligned}$$

with the help of the lemma? With the definition of the multiplier, we arrive at

$$m(y) = \gamma_k \frac{P_k(y)}{|y|^k}.$$

□

Theorem 6.9. *The classes of transformation defined by*

$$T(f) = c \cdot f + \lim_{\epsilon \rightarrow 0} \int_{|y| \geq \epsilon} \frac{\Omega(y)}{|y|^n} f(x - y) dy \quad \text{where } \Omega \text{ is smooth} \quad (6.4)$$

$$\widehat{Tf}(y) = m(y) \hat{f}(y) \quad \text{where } m \text{ is smooth} \quad (6.5)$$

are identical.

Proof. ($\implies$) Suppose that T takes the first form. We already showed that

$$m(x) = c + \int_{\mathbb{S}^{n-1}} \Gamma(x \cdot y) \Omega(y) d\sigma(y).$$

We can express m in terms of the spherical harmonics

$$\Omega(y) = \sum_{k=1}^{\infty} Y_k(y), \quad m(x) = \sum_{k=0}^{\infty} \tilde{Y}_k(X).$$

The previous theorem tells us that the ratios of spherical harmonics are explicit constants

$$\tilde{Y}_k(x) = \gamma_k Y_k(x).$$

For $N \neq M$, we get

$$\sup_{x \in \mathbb{S}^{n-1}} |m_M(x) - m_N(x)| \leq \sup_x \|\Gamma(x \cdot y)\|_{L_y^2(\mathbb{S}^{n-1})} \|\Omega_M - \Omega_N\|_{L_y^2(\mathbb{S}^{n-1})} \rightarrow 0$$

as $N, N \rightarrow 0$? First term in the product is bounded because

$$\Gamma(t) = \frac{\pi i}{2} \text{sgn}(t) + \log \frac{1}{|t|}$$

implies that

$$\sup_x \|\Gamma(x \cdot y)\|_{L_y^2(\mathbb{S}^{n-1})} \leq c_1 + c_2 \int_0^\pi |\log |\cos \theta||^2 (\sin \theta)^{n-2} d\theta < \infty.$$

So the sequence is Cauchy.

The smoothness of Ω allows us to meet condition of (6.1).

($\Leftarrow$) Suppose $m(x)$ is smooth on the unit sphere and set its spherical harmonics as above. Take

$$c = \tilde{Y}_0 \text{ and } Y_k(x) = \frac{1}{\gamma_k} \tilde{Y}_k(x).$$

Then Ω is infinitely differentiable. □

7 The Littlewood-Paley g -function

Definition 7.1. Let $f \in L^p(\mathbb{R}^n)$. We write $u(x, y)$ for its Poisson integral

$$u(x, y) = \int_{\mathbb{R}^n} P_y(t) f(x - t) dt.$$

Then we define $g(f)$ by

$$g(f)(x) = \left(\int_0^\infty |\nabla u(x, y)|^2 y dy \right)^{1/2}.$$

Theorem 7.2. Suppose $f \in L^p(\mathbb{R}^n)$, $1 < p < \infty$. Then $g(f)(x) \in L^p(\mathbb{R}^n)$ with

$$A'_p \|f\|_p \leq \|g(f)\|_p \leq A_p \|f\|_p.$$

Proof of $p = 2$ case. We have

$$\|g(f)\|_2^2 = \iint y |\nabla u(x, y)|^2 dx dy.$$

The formula for the Poisson integral

$$u(x, y) = \int \hat{f}(t) e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt.$$

We get

$$\begin{aligned} \frac{\partial u}{\partial y} &= \int -2\pi |t| \hat{f}(t) e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt, \\ \frac{\partial u}{\partial x} &= \int -2\pi i t_j \hat{f}(t) e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt. \end{aligned}$$

Together, we get

$$\int |\nabla u(x, y)|^2 dx = \int 8\pi^2 |t|^2 |\hat{f}(t)|^2 e^{-4\pi |t| y} dt.$$

Applying Fubini's, we have

$$\begin{aligned} \|g(f)\|_2^2 &= \int \left(\int |\nabla u(x, y)|^2 dx \right) y dy \\ &= \int \left(\int 8\pi^2 |t|^2 |\hat{f}(t)|^2 e^{-4\pi |t| y} dt \right) y dy \\ &= \int \left(\int 8\pi^2 |t|^2 y e^{-4\pi |t| y} dy \right) |\hat{f}(t)|^2 dt, \quad (\text{the constant is } \Gamma(2)) \end{aligned}$$

which implies

$$\|g(f)\|_2 = 2^{-1/2} \|f\|_2.$$

□

Remark 7.3. If we introduce

$$g_1(f)(x) = \left(\int_0^\infty \left| \frac{\partial u}{\partial y} \right|^2 y dy \right)^{1/2},$$

$$g_x(f)(x) = \left(\int_0^\infty |\nabla_x u(x, y)|^2 y dy \right)^{1/2},$$

then $g^2 = g_1^2 + g_x^2$ and we actually showed that

$$\|g_1(f)\|_2 = \|g_x(f)\|_2 = \frac{1}{2} \|f\|_2.$$

Proof when $p \neq 2$. (second inequality) When $p \neq 2$, we consider the Hilbert spaces $\mathcal{H}_1 = \mathbb{R}$ and

$$\mathcal{H}_2^0 = \left\{ f : |f|^2 = \int_0^\infty |f(y)|^2 y dy < \infty \right\},$$

$$\mathcal{H}_2 = \bigoplus_{i=1}^{n+1} \mathcal{H}_2^0.$$

Recall the definition and explicit expression of the Poisson kernel

$$P_y(x) = \int_{\mathbb{R}^n} e^{-2\pi i t \cdot x} e^{-2\pi |t| y} dt = \frac{c_n y}{(|x|^2 + y^2)^{\frac{n+1}{2}}}$$

We define

$$K_\epsilon(x) = \left(\frac{\partial P_{y+\epsilon}(x)}{\partial y}, \frac{\partial P_{y+\epsilon}(x)}{\partial x_1}, \dots, \frac{\partial P_{y+\epsilon}(x)}{\partial x_k} \right).$$

For each x , we have $K_\epsilon(x) \in \mathcal{H}^2$ from the explicit formula for the Poisson kernel. In particular, we have

$$\left| \frac{\partial P_y}{\partial y} \right|, \left| \frac{\partial P_y}{\partial x} \right| \leq \frac{A}{(|x|^2 + y^2)^{\frac{n+1}{2}}}??$$

Then for fixed x ,

$$\|K_\epsilon(x)\|_{\mathcal{H}_{2,y}}^2 \leq A^2(n+1) \int_0^\infty \frac{y dy}{(|x|^2 + (y+\epsilon)^2)^{n+1}} \leq A|x|^{-2n},$$

which means $\|K_\epsilon(x)\|_{\mathcal{H}_{2,y}} \in L^2(\mathbb{R}^n)$.

A similar estimate yields

$$\left\| \frac{\partial K_\epsilon(x)}{\partial x_j} \right\|_{\mathcal{H}_{2,y}}^2 \leq A \int_0^\infty \frac{y dy}{(|x|^2 + (y+\epsilon)^2)^{n+2}} \leq \frac{A}{|x|^{2n+2}}$$

We consider the operator T_ϵ defined by

$$T_\epsilon(f)(x) = \int K_\epsilon(t) f(x-t) dt$$

Observe that

$$|T_\epsilon(f)(x)| = \left(\int_0^\infty |\nabla u(x, y+\epsilon)|^2 y dy \right)^{1/2} \leq g(f)(x).$$

Then $\|T_\epsilon f(x)\|_2 \leq 2^{-1/2} \|f\|_2$, which means $|\hat{K}_\epsilon(x)| \leq 2^{-1/2}$.

Then apply the Calderón-Zygmund theorem for Hilbert spaces.

(first inequality) Applying polarization to the identity

$$\|g_1(f)\|_2 = \frac{1}{2} \|f\|_2,$$

we have

$$\begin{aligned} \int f_1 \overline{f_2} dx &= 4 \int_{\mathbb{R}^n} \int_0^\infty y \frac{\partial u_1}{\partial y} \overline{\frac{\partial u_2}{\partial y}} dy dx \\ &\leq 4 \int g_1(f_1) g_1(f_2) dx \\ &\leq 4 \|g_1(f_1)\|_p \|g_1(f_2)\|_q \\ &\leq 4 A_q \|g_1(f_1)\|_p \end{aligned}$$

with $\|f_2\|_q \leq 1$.

□

7.1 Positive function

Lemma 7.4. *Suppose u is harmonic and strictly positive. Then*

$$\Delta(u)^p = p(p-1)u^{p-2}|\nabla u|^2.$$

Lemma 7.5. *Suppose $F(x, y)$ is continuous in $\overline{\mathbb{R}_+^{n+1}}$, of class C^2 in $\mathbb{R}_+^{n+1}$, and suitably small at infinity. Then*

$$\int_{\mathbb{R}_+^{n+1}} y \Delta F(x, y) dx dy = \int_{\mathbb{R}^n} F(x, 0) dx.$$

Proof. Green's theorem asserts

$$\int_D (y \Delta F(x, y)) dx dy = \int_{\partial D} \left(y \frac{\partial F}{\partial \nu} - F \frac{\partial y}{\partial \nu} \right) d\sigma,$$

where $D = B_r \cap \mathbb{R}_+^{n+1}$. We observe that the spherical part of the boundary of D vanishes as $r \rightarrow \infty$ under suitable decay conditions for F , namely

$$|F| \leq \frac{C}{(|x| + y)^{n+\epsilon}}, \quad |\nabla F| \leq \frac{C}{(|x| + y)^{n+1+\epsilon}}.$$

□

Definition 7.6. We define the positive function g_λ^* as

$$(g_\lambda^*(f)(x))^2 = \int_0^\infty \int_{\mathbb{R}^n} \left(\frac{y}{|t| + y} \right)^{\lambda n} |\nabla u(x - t, y)|^2 y^{1-n} dt dy.$$

Definition 7.7. Let Γ be a fixed proper cone in $\mathbb{R}_+^{n+1}$ which has a vertex at the origin and contains $(0, 1)$. We may take

$$\Gamma = \{(t, y) \in \mathbb{R}_+^{n+1} : |t| < y, y > 0\}$$

and let $\Gamma(x)$ denote the translated cone. We define the positive function $S(f)(x)$ by

$$\begin{aligned} [S(f)(x)]^2 &= \int_{\Gamma(x)} |\nabla u(t, y)|^2 y^{1-n} dy dt \\ &= \int_{\Gamma} |\nabla u(x - t, y)|^2 y^{1-n} dy dt \end{aligned}$$

Proposition 7.8. *We assert that*

$$g(f)(x) \leq CS(f)(x) \leq C_\lambda g_\lambda^*(f)(x).$$

Proof. For the second inequality, check that within the cone we have

$$|t| < y \implies |t| + y < 2y \implies \frac{y}{|t| + y} > \frac{1}{2}.$$

For the first inequality we just want to show that $g(f)(0) \leq CS(f)(0)$. Let B_y be the ball in $\mathbb{R}_+^{n+1}$ centered at $(0, y)$ and tangent to the boundary of the cone Γ (in some sense this is the maximal ball that is still contained in the upper half plane). The radius of B_y is proportional to y . The partial derivatives of u are also harmonic functions, and obey the mean value property

$$\begin{aligned} \frac{\partial u}{\partial y}(0, y) &= \frac{1}{|B_y|} \int_{B_y} \frac{\partial u}{\partial y}(x, s) dx ds, \\ \implies \left| \frac{\partial u}{\partial y}(0, y) \right|^2 &= \frac{1}{|B_y|} \int_{B_y} \left| \frac{\partial u}{\partial y}(x, s) \right|^2 dx ds, \end{aligned}$$

by Jensen's inequality. Now multiply by y and integrate with respect to y for

$$\begin{aligned} \int_0^\infty y \left| \frac{\partial u(0, y)}{\partial y} \right|^2 dy &\leq C \int_0^\infty y^{-n} \int_{B_y} \left| \frac{\partial u}{\partial y}(x, s) \right|^2 dx ds dy \\ &\leq C \int_\Gamma y^{1-n} \left| \frac{\partial u}{\partial y}(x, y) \right|^2 dx dy, \end{aligned}$$

because $(x, s) \in B_y$ implies that y is comparable to s . Now repeat for the remaining partial derivatives. $\square$

Theorem 7.9. *Let λ be a parameter which is greater than 1. Suppose $f \in L^p(\mathbb{R}^n)$. Then*

1. *For every $x \in \mathbb{R}^n$, $g(f)(x) \leq C_\lambda g_\lambda^*(f)(x)$.*
2. *If $1 < p < \infty$, and $p > 2/\lambda$, then*

$$\|g_\lambda^*(f)\|_p \leq A_{p,\lambda} \|f\|_p.$$

Definition 7.10. Let $\mu \geq 1$, and write

$$M_\mu(f)(x) = \left(\sup_{r>0} \frac{1}{m(B(x, r))} \int_{B(x, r)} |f(y)|^\mu dy \right)^{1/\mu}.$$

Check that for $p > \mu$, we have

$$\|M_\mu(f)\|_p \leq A_{p,\mu} \|f\|_p.$$

Lemma 7.11. *Let $f \in L^p(\mathbb{R}^n)$ for $p \geq \mu \geq 11$. If u is the Poisson integral of f , then*

$$|u(x - t, y)| \leq A \left(1 + \frac{|t|}{y} \right)^n M(f)(x), \quad (7.1)$$

$$|u(x - t, y)| \leq A_\mu \left(1 + \frac{|t|}{y} \right)^{n/\mu} M_\mu(f)(x). \quad (7.2)$$

Proof of lemma. The inequality is unchanged under dilation $(x, t, y) \mapsto (\delta x, \delta t, \delta y)$, so we only need to consider $y = 1$.

We have

$$|u(x - t, 1)| = f(x) * P_1(x - t), \quad P_1(x) = \frac{c_n}{(1 + |x|^2)^{\frac{n+1}{2}}}.$$

[Theorem 5.4](#) tells us that

$$|u(x - t, 1)| \leq A_t(Mf)(x), \quad A_t = \int Q_t(x) dx,$$

where $Q_t(x)$ is the smallest decreasing radial majorant of $P_1(x - t)$, given by

$$Q_t(x) = c_n \cdot \sup_{|x'| \geq |x|} \left(\frac{1}{(1 + |x' - t|^2)^{(n+1)/2}} \right).$$

We have the following estimates

$$\begin{aligned} Q_t(x) &\leq c_n & |x| &\leq 2|t|, \\ Q_t(x) &\leq A'(1 + |x|^2)^{-\frac{n+1}{2}} & |x| &\geq 2|t|, \end{aligned}$$

so $A_t \leq A(1 + |t|^n)$ gives us (7.1).

To raise to the μ th power, observe that

$$\begin{aligned} u(x - t, y) &= \int P_y(s) f(x - t - s) ds, \\ |u(x - t, y)|^\mu &\leq \int P_y(s) |f(x - t - s)|^\mu ds = U(x - t, y), \end{aligned}$$

where U is the Poisson integral of $|f|^\mu$. So we can apply (7.1) to U for

$$|u(x - t, y)| \leq A^{1/\mu} (1 + |t|/y)^{n/\mu} (M(|f|^\mu))(x)^{1/\mu}.$$

□

Proof. ($p \geq 2$ case) Let ψ be positive function on $\mathbb{R}^n$. Then

$$\int_{\mathbb{R}^n} (g_\lambda^*(f))(x)^2 \psi(x) dx \leq A_\lambda \int_{\mathbb{R}^n} (g(f)(x))^2 (M\psi)(x) dx.$$

The left hand side equals

$$\begin{aligned} &\int_{\mathbb{R}^n} \int_0^\infty \int_{\mathbb{R}^n} \left(\frac{y}{|t| + y} \right)^{\lambda n} |\nabla u(x - t, y)|^2 y^{1-n} \psi(x) dt dy dx \\ &= \int_0^\infty \int_{\mathbb{R}^n} \left[y^{-n} \int_{\mathbb{R}^n} \left(\frac{y}{|t - x| + y} \right)^{\lambda n} \psi(x) dx \right] |\nabla u(t, y)|^2 y dt dy, \end{aligned}$$

so we can apply Theorem 5.4 with $\epsilon := y$.

In the case that $p > 2$, set $1/q + 2/p = 1$, and take the supremum on the left hand side over all $\|\psi\|_q \leq 1$. Then apply Hölder duality, inequalities for the g -function and the boundedness of the maximal function operator.

For $p < 2$, we need the restriction that $p > 2/\lambda$. We can find μ close to p with $1 \leq \mu < p$ such that

$$\lambda' = \lambda - \frac{2 - p}{\mu} > 1,$$

Then apply the lemma for

$$|u(x-t, y)| \left(\frac{y}{y+|t|} \right)^{n/\mu} \leq AM_\mu(f)(x).$$

We have

$$\begin{aligned} (g_\lambda^*(f)(x))^2 &= \frac{1}{p(p-1)} \int_{\mathbb{R}_+^{n+1}} y^{1-n} \left(\frac{y}{y+|t|} \right)^{\lambda n} u^{2-p} |\Delta u^p| dt dy \\ &\leq A^{2-p} (M_\mu(f)(x))^{2-p} I^*(x), \end{aligned}$$

where

$$I^*(x) = \int_{\mathbb{R}_+^{n+1}} y^{1-n} \left(\frac{y}{y+|t|} \right)^{\lambda' n} \Delta u^p(x-t, y) dt dy.$$

We check that

$$\begin{aligned} \int_{\mathbb{R}^n} I^*(x) dx &= \int_{\mathbb{R}_+^{n+1}} \int_{x \in \mathbb{R}^n} y^{1-n} \left(\frac{y}{y+|t-x|} \right)^{\lambda' n} \Delta u^p(t, y) dx dt dy \\ &= C_{\lambda'} \int_{\mathbb{R}_+^{n+1}} y \Delta u^p(t, y) dt dy = C_{\lambda'} \int_{\mathbb{R}^n} u^p(t, 0) dt \leq C_{\lambda'} \|f\|_p^p \end{aligned}$$

by a change of variables $x \mapsto xy$. □

8 Multipliers

Theorem 8.1. *Suppose that $m \in C^k(\mathbb{R}^n \setminus \{0\})$, where $k > n/2$. Assume that for all $|\alpha| \leq k$, we have*

$$\left| \left(\frac{\partial}{\partial x} \right)^\alpha m(x) \right| \leq B|x|^{-|\alpha|}.$$

Then $m \in \mathcal{M}_p$ for all $1 < p < \infty$.

9 Partial sum operators

Definition 9.1. Let ρ be a rectangle in $\mathbb{R}^n$, by which we mean that the sides are parallel to the axes. We define the associated “partial sum operator” by

$$S_\rho(f)^\wedge = \mathbf{1}_\rho \cdot \hat{f}, \quad f \in L^2 \cap L^p(\mathbb{R}^n).$$

Theorem 9.2. *For $1 < p < \infty$, $f \in L^2 \cap L^p$, we have*

$$\|S_\rho(f)\|_p \leq A_p \|f\|_p,$$

where the constant A_p is independent of the rectangle ρ .

Definition 9.3. Let $\mathcal{R} = \{\rho_j\}_{j=1}^\infty$ be a sequence of rectangles. We define the operator $S_{\mathcal{R}} : L^2(\mathbb{R}^n, \ell^2) \rightarrow L^2(\mathbb{R}^n, \ell^2)$, given by

$$S_{\mathcal{R}}((f_1, f_2, \dots)) = (S_{\rho_1}(f_1), \dots, S_{\rho_j}(f_j), \dots).$$

We get the following generalized theorem

Theorem 9.4. *Let $f \in L^2(\mathbb{R}^n, \ell^2) \cap L^p(\mathbb{R}^n, \ell^2)$. Then for $1 < p < \infty$,*

$$\|S_{\mathcal{R}}(f)\|_p \leq A_p \|f\|_p,$$

where A_p is independent of the family of rectangles $\mathcal{R}$.

10 Dyadic decomposition

Let Δ denote the dyadic decomposition of $\mathbb{R}^n$. In the sense of L^2 convergence, we expect

$$\sum_{\rho \in \Delta} s_\rho = \text{id}.$$

Because the blocks are mutually orthogonal, we actually have the stronger statement

$$\sum_{\rho \in \Delta} \|S_\rho f\|_2^2 = \|f\|_2^2.$$

Theorem 10.1. *Suppose $f \in L^p(\mathbb{R}^n)$, $1 < p < \infty$. Then $(\sum_{\rho \in \Delta} |S_\rho f(x)|^2)^{1/2} \in L^p(\mathbb{R}^n)$ and in fact*

$$\left\| \left(\sum_{\rho \in \Delta} |S_\rho f(x)|^2 \right)^{1/2} \right\|_p \quad \text{and} \quad \|f\|_p$$

are comparable.